

Yusuf Abdulle

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Education

- King's College London**, PhD, Machine Learning for Health 2024–2028
London, UK
- Thesis: *Multimodal Foundation Models for Rare Neurological Disease: Knowledge Graphs, Genomics, and Clinical Trajectories in Motor Neuron Disease*
- University College London**, MSc Clinical Neuroscience – Merit 2020–2021
London, UK
- University of Brighton**, BSc (Hons) Biomedical Science – First Class Honours 2017–2020
Brighton, UK

Research Experience

- Intern, InnovationLab | *Relation Therapeutics*** Aug 2026–Nov 2026
London, UK
- Developing computational pipelines for spatial transcriptomics and biophysical modelling to support internal drug discovery programmes.
 - Building scalable workflows to integrate and analyse spatial and molecular data for therapeutic research.
- Research Assistant | *University College London*** Sep 2023–Sep 2024
London, UK
- Developed and benchmarked vision-language and large language models for healthcare applications, with findings disseminated at ACL 2024 and IJCAI 2024.
 - Developed quantitative frameworks to measure AI-induced health disparities across race, sex, age, and socioeconomic status.
 - Mentored undergraduate researchers and research interns in project design, methodology, and dissemination.
- Translational Data Scientist | *BioCortex*** Oct 2022–Apr 2023
London, UK
- Developed first-principles computational simulations to model clinical trial failure modes associated with the gut microbiome and host-microbiome interactions.
 - Built automated DNA sequence benchmarking pipelines against curated reference databases for model evaluation and optimisation.
- Research Assistant | *University College London*** Sep 2021–Oct 2022
London, UK
- Developed Graph Neural Networks for MRI-based detection of Frontotemporal Dementia and ALS using PyTorch Geometric and Deep Graph Library.
 - Built AWS (EC2/S3) pipelines for high-throughput processing of 3T MRI datasets and stochastic models for healthcare pathway optimisation.

Publications

Under Review / Preprints

- Abdulle, Y.**, et al. “Symptom-based phenotype discovery in motor neuron disease using natural language processing of electronic health records.” *medRxiv* (2026).
- Dong, H., Vasu, R., **Abdulle, Y.**, et al. “A Large Language Model based Framework for Dementia Related Hypothesis Generation.” (2026).
- Abdulle, Y.**[^], Wu, J.[^], Budhdeo, S., et al. “Characteristics and Early Diagnosis of Motor Neuron Disease (MND) in 67 million individuals in England: A comparative study on phenotyping models derived by AI, Knowledge Graphs and the MND Association.” *medRxiv* (2025).
- Budhdeo, S., Zhang, J., **Abdulle, Y.**, et al. “Scoping review of knowledge graph applications in biomedical and healthcare sciences.” *Wellcome Open Research* (2025).

Journal Articles

- Spears, S. D. J., **Abdulle, Y. F.**, Lester, T., et al. “Understanding neck collar preferences and user experiences in motor neuron disease: A survey-based study.” *Disability and Health Journal* (2024).
- Spears, S. D. J., **Abdulle, Y. F.**, Korovilas, D., et al. “Neck collar assessment for people living with motor neuron disease: Are current outcome measures suitable?” *Interactive Journal of Medical Research* (2023).

Conference Papers

- Kim, Y., **Abdulle, Y.**, Wu, H. “BioHopR: A Benchmark for Multi-Hop, Multi-Answer Reasoning in Biomedical Domain.” *Findings of ACL* (2025).
- Kim, Y., Wu, J., **Abdulle, Y.**, Wu, H. “MedExQA: Medical Question Answering Benchmark with Multiple Explanations.” *BioNLP Workshop, ACL* (2024).
- Kim, Y., Wu, J., **Abdulle, Y.**, Gao, Y., Wu, H. “Human-in-the-Loop Chest X-Ray Diagnosis: Enhancing Large Multimodal Models with Eye Fixation Inputs.” *TAI4H, Springer LNCS* (2024).
- Kim, Y., Wu, J., **Abdulle, Y.**, Gao, Y., Wu, H. “Enhancing Human-Computer Interaction in Chest X-Ray Analysis Using Vision and Language Models with Eye Gaze Patterns.” *MICCAI* (2024).
- Groves, E., Wang, M., **Abdulle, Y.**, et al. “Benchmarking and Analyzing In-context Learning, Fine-tuning and Supervised Learning for Biomedical Knowledge Curation.” *VLDB Workshops* (2023).

Conference Abstracts

- Budhdeo, S., **Abdulle, Y.**, Sharma, N., Cosco, T. “Meta-analysis of cerebrospinal fluid immune markers in frontotemporal dementia patients compared to healthy controls.” *European Journal of Neurology* (2023).
- Budhdeo, S., **Abdulle, Y.**, Kaczmarczyk, I., Sharma, N. “Using a PET atlas to probe neurotransmitter-disease associations in Mild Cognitive Impairment and Alzheimer’s disease.” *European Journal of Neurology* (2023).

Honors & Awards

- **Longitude Prize on ALS – Discovery Award – 2026**; selected as one of 20 international teams from nearly 100 applications.
- **EPSRC Centre for Doctoral Training in Data-Driven Health (DRIVE-Health) Fellowship – 2024.**
- **Santander Summer Studentship – 2019.**

Presentations

Conference Presentations & Posters

- **International Symposium for ALS/MND 2025**, San Diego: “Characteristics and early diagnosis of MND in 67 million individuals in England.”
- **ACL 2025**, Vienna: “BioHopR: A Benchmark for Multi-Hop, Multi-Answer Reasoning in Biomedical Domain.”
- **HealTAC 2025**, Glasgow: “Can GPT-4 be a good red flagger for MND?”
- **HealTAC 2025**, Glasgow: “A Large Language Model based Framework for Dementia Related Hypothesis Generation.”
- **ENCALS 2025**, Turin: “Early ALS Phenotyping and COVID-19 Survival Analysis Using EHR Data.”
- **MICCAI 2024**, Marrakesh: “Enhancing Human-Computer Interaction in Chest X-Ray Analysis Using Vision and Language Models with Eye Gaze Patterns.”
- **MICCAI 2024**, Marrakesh: Organiser, Foundation Models for Medical Imaging Workshop (FOMMIA).
- **TAI4H 2024**, Jeju Island: “Human-in-the-Loop Chest X-Ray Diagnosis: Enhancing Large Multimodal Models with Eye Fixation Inputs.”
- **BioNLP Workshop 2024**, Bangkok: “MedExQA: Medical Question Answering Benchmark with Multiple Explanations.”
- **OHBM 2023**, Vancouver: “Using a PET neurotransmitter atlas to probe associations in Alzheimer’s disease and MCI.”
- **EAN Congress 2023**, Budapest: “Using a PET atlas to probe neurotransmitter-disease associations in Mild Cognitive Impairment and Alzheimer’s disease.”

Academic Service

- **Workshop Organiser:** Foundation Models for Medical Imaging (FOMMIA), MICCAI 2024.
- **Program Committee:** Multimodal Large Language Models in Clinical Practice (MLLMCP), MICCAI 2025.
- **Reviewer:** IEEE International Conference on Healthcare Informatics (ICHI), 2026.

Teaching & Tutorials

- **Graduate Teaching Assistant**, Department of Biostatistics and Health Informatics, King’s College London: Introduction to Health Informatics (2024–2026); Machine Learning for Health and Bioinformatics (2024–2026).
- **Tutorial Instructor:** “Multimodal Large Language Models for Healthcare,” University of Glasgow (2025).

Technical Skills

- **Machine Learning:** Deep Learning, Graph Neural Networks, Knowledge Graphs, PyTorch, PyTorch Geometric, TensorFlow, Scikit-Learn.
- **Programming & Data:** Python, R, Bash, NumPy, Pandas, Statistical Modelling, NetworkX.
- **Infrastructure:** AWS (EC2, S3), GCP, Docker, Git, PostgreSQL.
- **Languages:** English, Somali.